



// THE QUESTION EVERY TRAINED MODEL HIDES

# IS ITS CONFIDENCE EARNED, OR HAS IT SIMPLY RUN OUT OF DATA?

FORGE flips a trained point estimate into a full Bayesian posterior, splits its predictive variance into the part the data cannot fix and the part the model has yet to learn, and reads the balance off one dial. Four channels then say whether that reading is earned.

## 01 BASE MEASURE

### THE BUDGET

$$\text{Var}(y|x) = \underbrace{E_\theta[\text{Var}(y|x,\theta)]}_{\text{A ALEATORIC}} + \underbrace{\text{Var}_\theta(E[y|x,\theta])}_{\text{E EPISTEMIC}}$$

#### A IRREDUCIBLE · LIVES IN THE DATA

Noise in the target itself. Average the prediction's variance over the posterior and this is what is left. It must not move with N – exactly what the A-invariance referee tests.

#### E REDUCIBLE · LIVES IN THE MODEL

Spread from not knowing the weights. It collapses as data pins the posterior down. This is the only part of the doubt that money, labels or events can buy back.

#### WHY VARIANCE, NOT ENTROPY

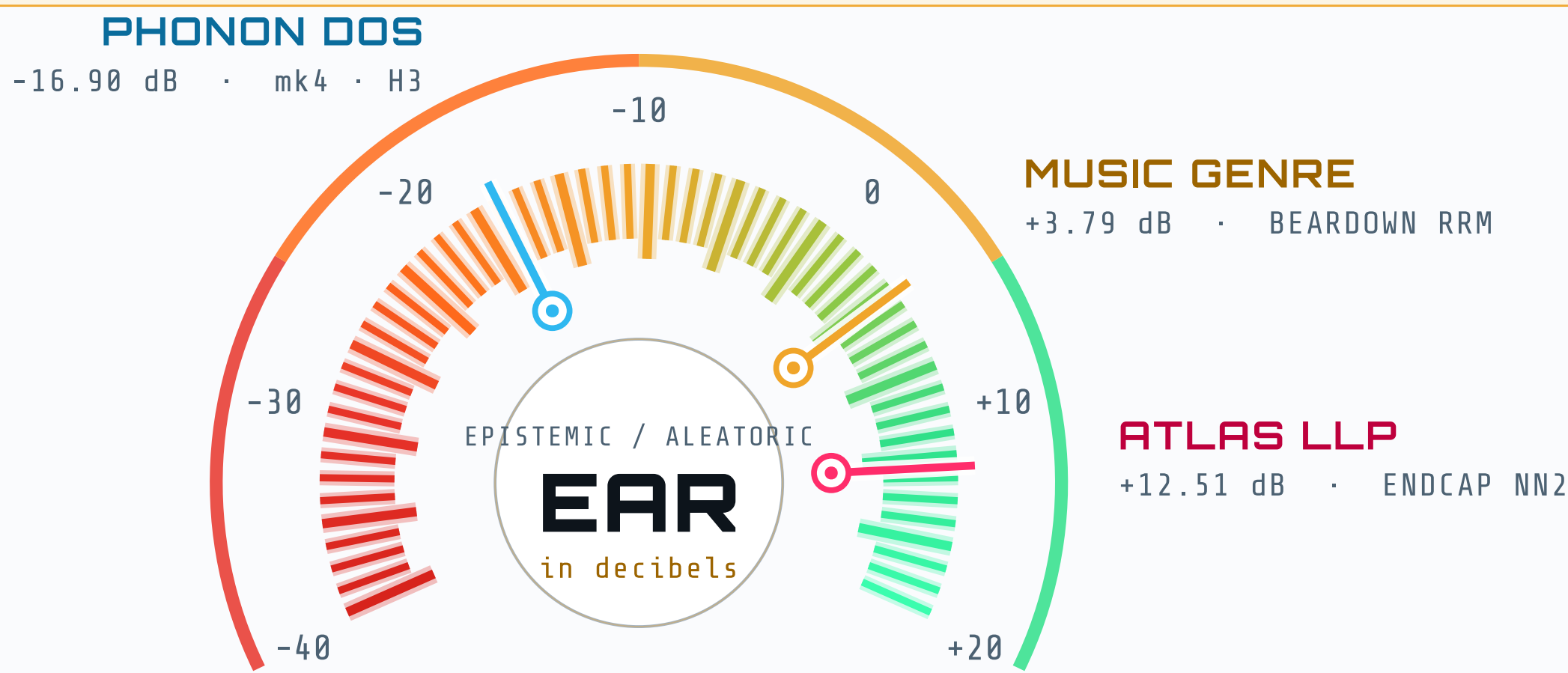
Differential entropy is not invariant under reparameterisation and can go negative; mutual information on a continuous DOS head needs a density FORGE would have to invent. Variance is defined for every head on the board, carries the units of the target, and its ratio is literally a signal-to-noise ratio – so it can be read in decibels.

$\tau$  is not tuned by hand – MacKay evidence, no held-out labels.

## 02 THE READING

### ONE DIAL, THREE VERDICTS

$$\text{EAR} = 10 \log_{10} \left( \frac{E}{A} \right) \text{ dB} \quad \text{reducibility} = E/(E+A)$$



|                |            |                                                  |
|----------------|------------|--------------------------------------------------|
| IGNORANCE-LED  | +5 to +20  | doubt is reducible; collecting data pays         |
| BALANCED       | -10 to +5  | E and A within an order – check prior leak first |
| NOISE-LEARNING | -25 to -10 | A dominates 10–300×; expect small returns        |
| NOISE-BOUND    | -40 to -25 | the model has taken all the signal there is      |

EAR is a field, not a scalar – it resolves along ten genre classes, an energy bin  $\omega$ , or an ABCD region.

## 03 MEASURED, NOT ASSUMED

### THE BOARD

| MODEL / BOARD                                   | d    | EAR dB | REDUCIBLE | PRIOR LEAK | VERDICT       |
|-------------------------------------------------|------|--------|-----------|------------|---------------|
| ATLAS LLP<br>endcap nn2 · fold 0                | 129  | +12.51 | 94.7 %    | –          | IGNORANCE-LED |
| ATLAS LLP<br>endcap nn1 · fold 0                | 129  | +5.04  | 76.1 %    | –          | IGNORANCE-LED |
| MUSIC GENRE<br>beardown cfg17                   | 384  | +4.17  | 72.3 %    | –          | BALANCED      |
| MUSIC GENRE<br>beardown RRM · dashboard default | 384  | +3.79  | 70.5 %    | –          | BALANCED      |
| MUSIC GENRE<br>beardown mk1                     | 512  | –0.11  | 49.4 %    | –          | BALANCED      |
| MUSIC GENRE<br>beardown 3-sec                   | 768  | –0.99  | 44.3 %    | 0.03 %     | BALANCED      |
| PHONON DOS<br>mk4 H3_planted_band               | 1632 | –16.90 | 2.00 %    | –0 at k*   | NOISE-BOUND   |

8 boards exported · 7 shown · one instrument, three answers

#### ATLAS – SPEND ON DATA.

95% of NN2's predictive variance is reducible. By region the story sharpens: signal-like quadrant A reads –2.63 dB while background corner D reads +17.83.

#### PHONON – SPEND ON THE MODEL.

Only 2.00% reducible. A larger spectrum library will not move this head; its doubt is the target's own spread.

#### GENRE – IT DEPENDS ON WHICH MODEL YOU BUILT.

Five boards on one dataset straddle parity, –0.99 to +4.17 dB. The dataset never changed.

## 04 THE FORGE FOUR

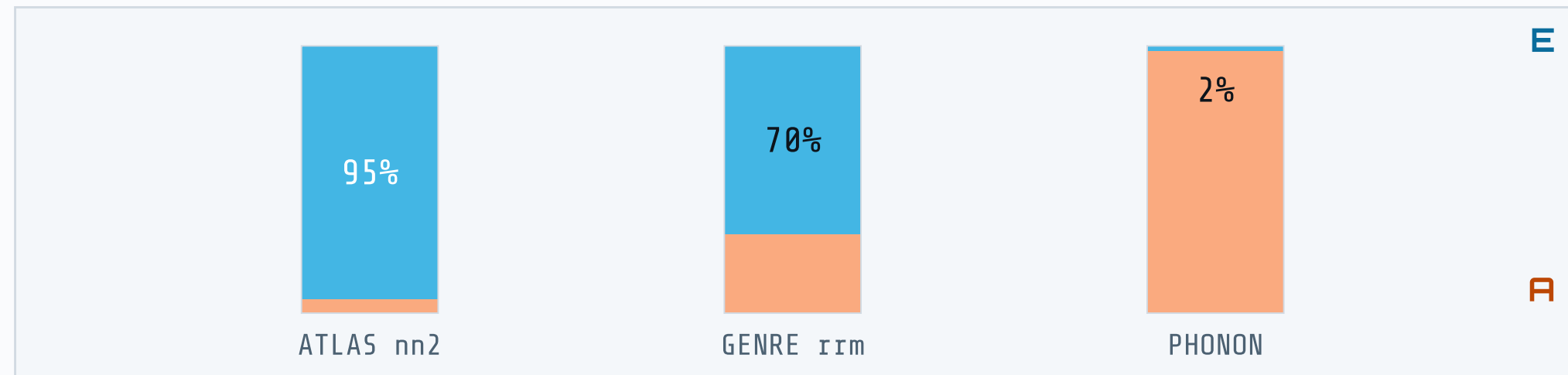
FOUR VIEWS OF ONE VARIANCE BUDGET

C1 splits the budget · C2 verifies the split · C3 sources it · C4 locates it  
together they say whether the dial's reading is earned – and where the doubt lives

### C1 SPLIT IT LLLA

$$\Sigma = (\Phi^T H \Phi + \tau I)^{-1}$$

Last-layer Laplace at the evidence optimum. One pass, no retraining, no held-out labels. Every row and every bin gets an A and an E.



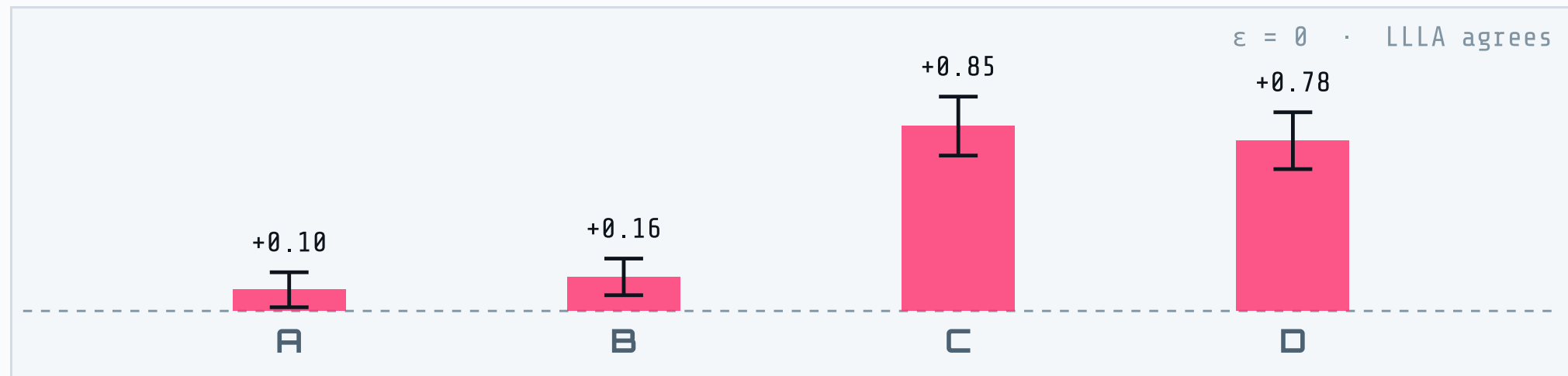
reducible share  $E/(E+A)$  · three headline boards

answers – how much of this doubt is mine?

### C2 VERIFY IT HMC · NUTS

$$\epsilon = E_{\text{HMC}} / E_{\text{LLLA}} - 1$$

The gold standard on the same head. Dispersed chains gated at R-hat < 1.01.  $\epsilon$  per bin, with MCSE, says whether the fast path told the truth.



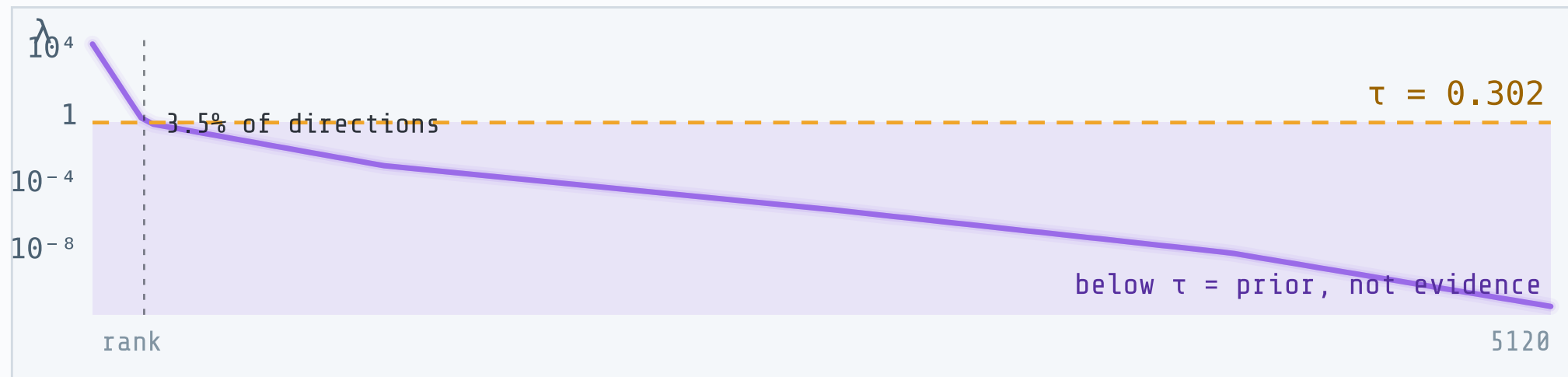
ATLAS nn1 ·  $\epsilon$  by ABCD quadrant,  $\pm 2$  MCSE

answers – can I trust the split?

### C3 SOURCE IT GGN EIGENSPECTRUM

$$\text{leak} = \Sigma(\lambda < \tau) E_\lambda / E$$

Directions whose data curvature falls below  $\tau$  carry prior, not evidence. Their share of E is the prior leak – the dial reading itself back to you.



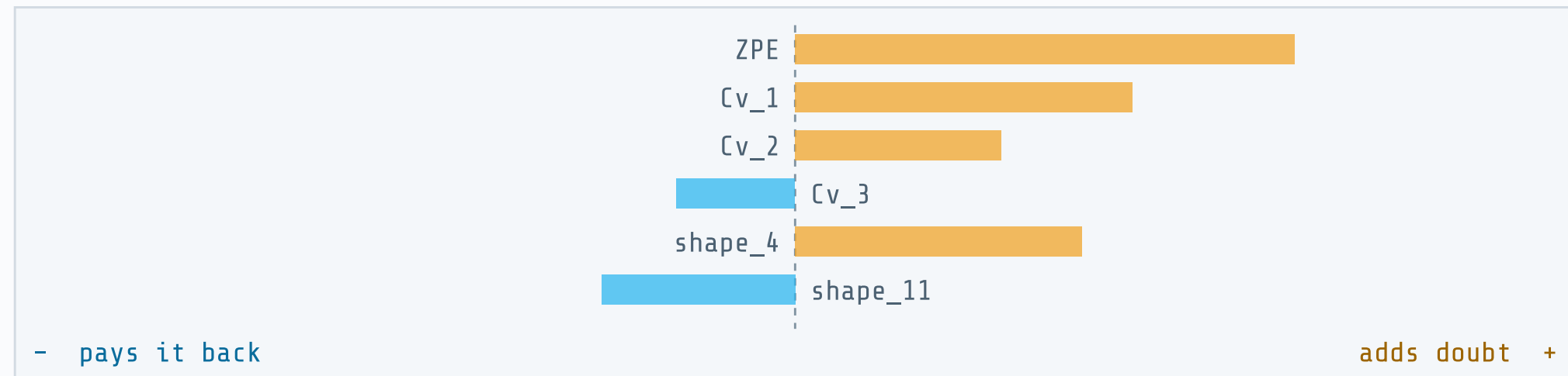
genre beardown · 5120 directions · 181 above  $\tau$

answers – is this doubt earned from data?

### C4 LOCATE IT PER-FEATURE POSTERIOR

$$a_k = \phi_k(\Sigma \phi)^{-1}_k, \quad \Sigma a_k = E$$

Euler allocation splits E across features exactly, and it is signed – a feature can pay doubt back instead of only adding it.

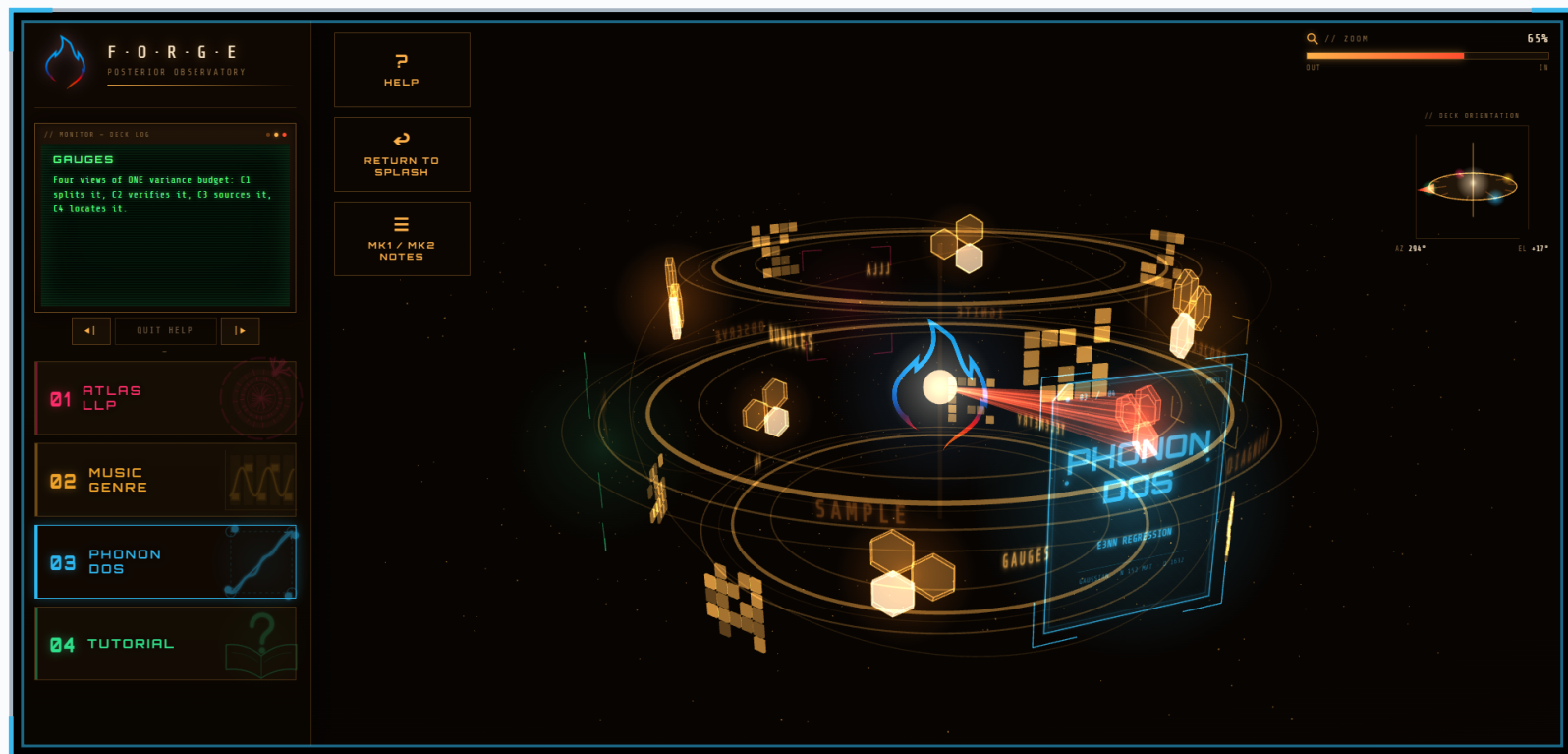


signed rows · phonon C4 rows are named physics

answers – which features are the doubt?

## 05 BUILT, NOT MOCKED

### THE INSTRUMENT



#### NAVIGATION DECK

Holographic model select, CRT deck log, locked per-model accents.



#### FORGE DASHBOARD

Segmented EAR dial, channel rail, metric detail, standing A/E defs.

## 06 FINDINGS

### WHAT THE INSTRUMENT CAUGHT

#### CALIBRATION IS NOT TRUST.

Head H7 posted PICIP95 = 0.951, dead on nominal, and still failed the A-invariance referee at 183.9%: it had absorbed model bias into the aleatoric term. A  $\sigma^2$  fitted to a model's own residuals always covers well – coverage cannot see that. The N-invariance test can.

#### THE FAST PATH IS OPTIMISTIC ABOUT ITS OWN DOUBT.

Two dispersed NUTS chains (R-hat 1.0005, ESS 1400) on the highlighted RRM board: LLLA over-states E by 8× to 66×, every bin beyond 2 MCSE. On ATLAS it agrees in the two high-score quadrants and over-states ~80% in the low ones. Over-cautious – and now quoted, not assumed.

#### THE FRONTIER IS THE RESULT.

Phonon rank 51: at  $k = 0$  the head reads frac@10% 0.717 with 0.01% reducible – accurate and blind. At  $k^* = -4.97$  it reads 0.664 with 2.5–3.9% and better calibration. Accuracy and epistemic signal trade; FORGE prices the trade rather than hiding it.

MEASURED FRONTIER · PHONON RANK 51

